6. Energy Management

Operating Systems (OS)
Master in Innovation and Research in Informatics (MIRI)
Facultat d'Informàtica de Barcelona (FIB)
Universitat Politècnica de Catalunya (UPC)
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FIB

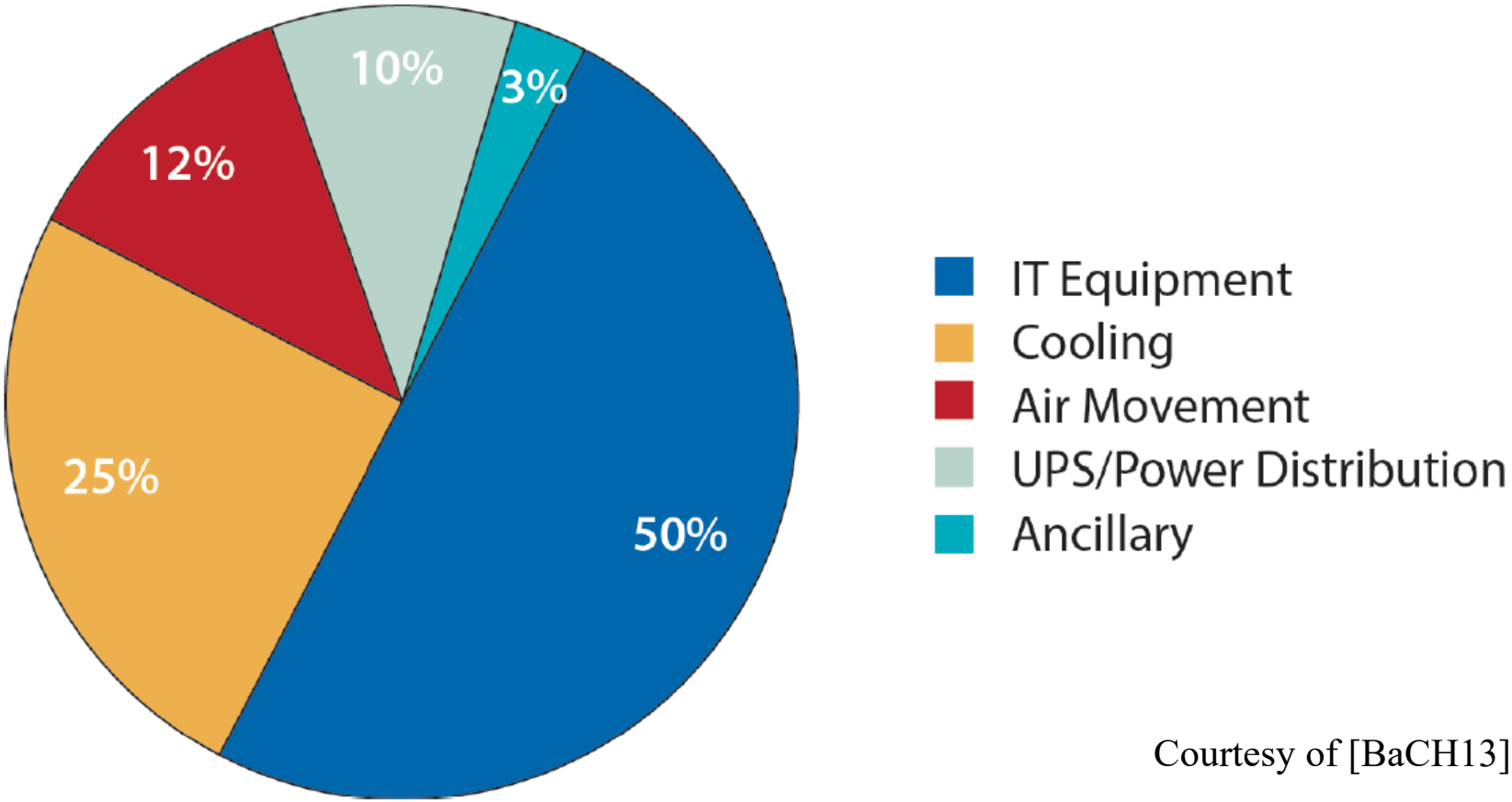
Motivation

- A consequence of the WSC architecture is an increment of energy-related costs
- This makes **energy efficiency** an important challenge for WSCs
- There are several subsystems in a WSC which have an impact in its energy efficiency
  - IT equipment
  - Power supply subsystem (*)
  - Cooling subsystem (*)

(*) Those systems are less relevant from an energy point of view in desktop systems

Motivation

- Power consumption in a traditional WSC (PUE=2)



Courtesy of [BaCH13]

Energy management strategy

1. Enable awareness of energy impact & carbon footprint
2. Increase the energy efficiency of the IT equipment
3. Increase the energy efficiency of the cooling and power supply subsystems
4. Increase the use of renewable energy
5. Exploit opportunities in the energy markets
6. Customize this strategy to the WSC reality

READING REPORT

[Guitart17] Guitart, J., *Toward Sustainable Data Centers: a Comprehensive Energy Management Strategy*, Computing, Vol. 99 (6), pp. 597-615, June 2017

Contents

- **Energy awareness**
- Efficiency of the IT equipment
- Cooling subsystem
- Power supply subsystem

Energy awareness

- Energy efficiency requires the WSC to be aware of its energy and carbon impact

1. Definition of metrics for energy efficiency
2. Energy consumption accounting
   a) Power metering: Measure the actual power consumption by instrumenting the hardware
   b) Power modelling: Estimate power consumption by means of correlated indirect evidences
      - Performance monitoring counters
      - Resource utilization metrics

Contents

- **Energy awareness**
  - **Metrics**
  - Power metering
  - Power modelling
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Energy efficiency

- Energy efficiency measures the energy used by a WSC to run a particular workload
- It depends on three independent factors:

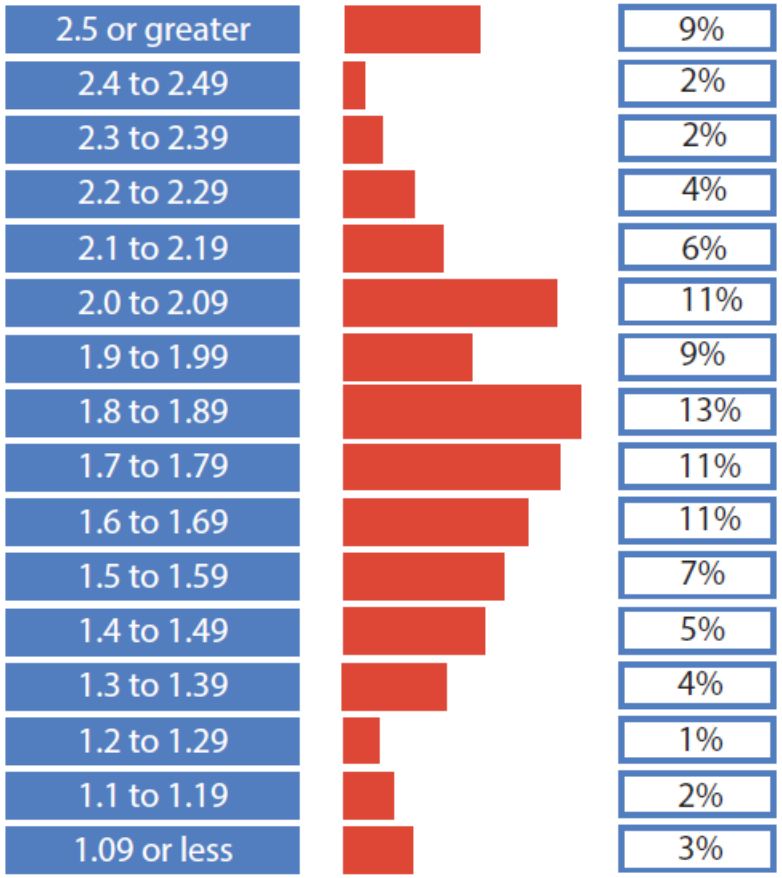
$$\text{Efficiency} = \frac{\text{Computation}}{\text{Total Energy}} = \underbrace{\left(\frac{1}{\text{PUE}}\right)}_{\text{(a)}} \times \underbrace{\left(\frac{1}{\text{SPUE}}\right)}_{\text{(b)}} \times \underbrace{\left(\frac{\text{Computation}}{\text{Total Energy to Electronic Components}}\right)}_{\text{(c)}}$$

a) measures the facility's efficiency
b) measures the server's power conversion efficiency
c) measures the server's architectural efficiency

Facility's efficiency: PUE

- Power Usage Effectiveness (PUE)
  - Captures the ratio of facility's total power to IT power
- e.g. 2016 LBNL's PUE report
  - Hyperscale WSCs (e.g. Google): 1.13
  - Traditional data centers: 1.6 – 2.35

2012 Uptime Institute PUE survey for 1100+ datacenters (mean PUE: 1.8 – 1.89)



Courtesy of [BaCH13]

Facility's efficiency: PUE

- Issues using PUE to compare WSCs
  1. Not all the PUE measurements include the same overheads

  ⇒ WSCs should adhere to the Green Grid guidelines in measurements and reporting

  2. Instantaneous PUEs differ from average PUEs

  ⇒ Annual averages reported from data aggregates are most useful for comparisons

  3. Some PUEs are based on optimal operating conditions and nominal performance values

  ⇒ PUE should be measured in real time

Server's power conversion efficiency

- SPUE accounts for overheads inside servers by measuring the ratio of server's total input power to its useful power
  - Useful power includes only the power consumed by the components directly involved in the computation: motherboard, disks, CPUs, DRAM …
  - Power may be lost in the server's power supply, voltage regulator modules, and cooling fans
- Today, Google's SPUE is around 1.11

Server's power conversion efficiency

- TPUE, defined as the product of PUE and SPUE, constitutes an accurate assessment of the end-to-end electromechanical efficiency of a WSC
- Today, Google's TPUE is around 1.23
- This means that the electromechanical components in a WSC have a limited potential for improvement

  $\Rightarrow$We need to improve the efficiency of computing!

Server's architectural efficiency

- Data Center energy Productivity (DCeP) accounts for how the energy delivered to electronic components is actually translated into **useful work**
  - It quantifies the useful work output that a WSC provides in relation to the amount of energy expended to produce it
  - Useful work in a WSC can take many forms
    - Difficult to compare different WSCs
  - Because of this, typically benchmarks have been used (e.g. SPECpower)

Contents

- **Energy awareness**
  - Metrics
  - **Power metering**
  - Power modelling
- Efficiency of the IT equipment
- Cooling subsystem
- Power supply subsystem

Power metering

1. Integrated metering solutions on the HW
   - Firmware offers an API to query the power usage

   a) Add specific power sensors
      - e.g. Dell PowerEdge, HP ProLiant

   b) Use software power models
      - e.g. Intel Running Average Power Limit (RAPL)

   ↑ Accurate power measurements for individual components

   ↓ Architecture dependent

   ↓ Intrusive method (metering procedure itself consumes power)

Power metering

2. Attach an external meter to measure AC power at the outlet
   a) General purpose meters (not appropriate for large-scale deployments)
      - e.g. PLOGG, Kill-A-Watt
   b) Intelligent Power Distribution Units (PDUs)
      - e.g. Raritan Dominion PX, HP Intelligent PDU

↑ Architecture independent

↑ Unobtrusive external measurements

↓ Can measure only system-level power

↓ Less accurate measurements

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  - Metrics
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Power modelling

- Power models can be used to account for energy consumption
  a) When direct measurement is not feasible
     - Performance overhead of measurements
     - Lack of scalability in large clusters
     - Cost of metering devices
  b) To assess the power consumption of individual software components such as processes or virtual machines
  c) To forecast the power consumption in the future or in hypothetical scenarios

Power modelling

- Models can focus on different aspects, but all of them are correlated
  a) Application workloads
  b) Servers power load and energy consumption
  c) Servers temperature and heat distribution
  d) Cooling and air flows
  e) Costs (operational and capital expenses)
- Models can work at different scales
  a) Time scales: msecs, secs, mins, hours
  b) Spatial scales: components, servers, racks, WSC

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- Energy awareness
- **Efficiency of the IT equipment**
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Efficiency of IT equipment

- Design **software** strategies for:

1. Smart use of power management features
   - Inactive low-power modes (a.k.a. turning-on/off)
   - Active low-power modes (a.k.a. frequency scaling)
2. Power-friendly resource allocation and scheduling of tasks
   - Temporal and spatial placement of tasks in:
     - The physical hosts of the WSCs
     - An ecosystem of WSCs
   - Elasticity on the use of resources of the tasks

Contents

- Energy awareness
- **Efficiency of the IT equipment**
  - **Low-power modes**
  - Resource allocation and scheduling
- Cooling subsystem
- Power supply subsystem

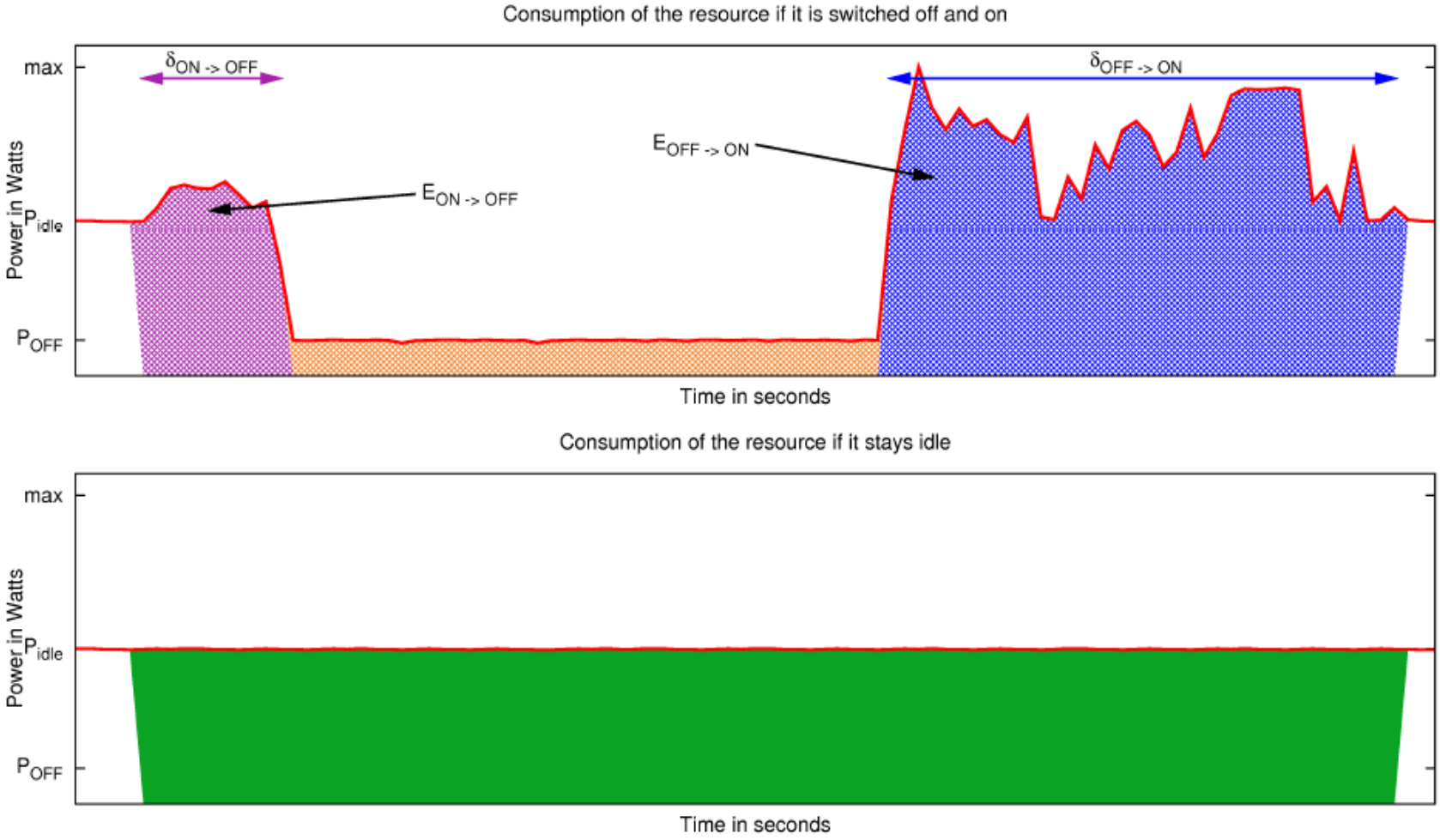
Inactive low-power modes

- Devices are not usable while in those modes, but do not consume energy (or minimal)
- IDEA: turn off hosts when they are idle and turn them on when they are needed again
  - Switching on/off takes time and consumes energy
  - Needs long idleness intervals (δ_{idle}) to be worth it
    - Not frequent in WSCs that use load distribution for performance and data replication for fault tolerance

$$P_{idle} \cdot \delta_{idle} > E_{ON \to OFF} + E_{OFF \to ON} + \left(P_{OFF} \cdot (\delta_{idle} - \delta_{ON \to OFF} - \delta_{OFF \to ON})\right)$$

$$\delta_{idle} > \frac{E_{ON \to OFF} + E_{OFF \to ON} - P_{OFF} \cdot (\delta_{ON \to OFF} + \delta_{OFF \to ON})}{P_{idle} - P_{OFF}}$$

Inactive low-power modes



Courtesy of [Orge11]

Active low-power modes

- DVFS: Dynamic Voltage & Frequency Scaling
  - Allows changing dynamically the operating voltage/frequency of the processor
- Observation: Dynamic power consumption of a CMOS circuit decreases quadratically with voltage and linearly with frequency

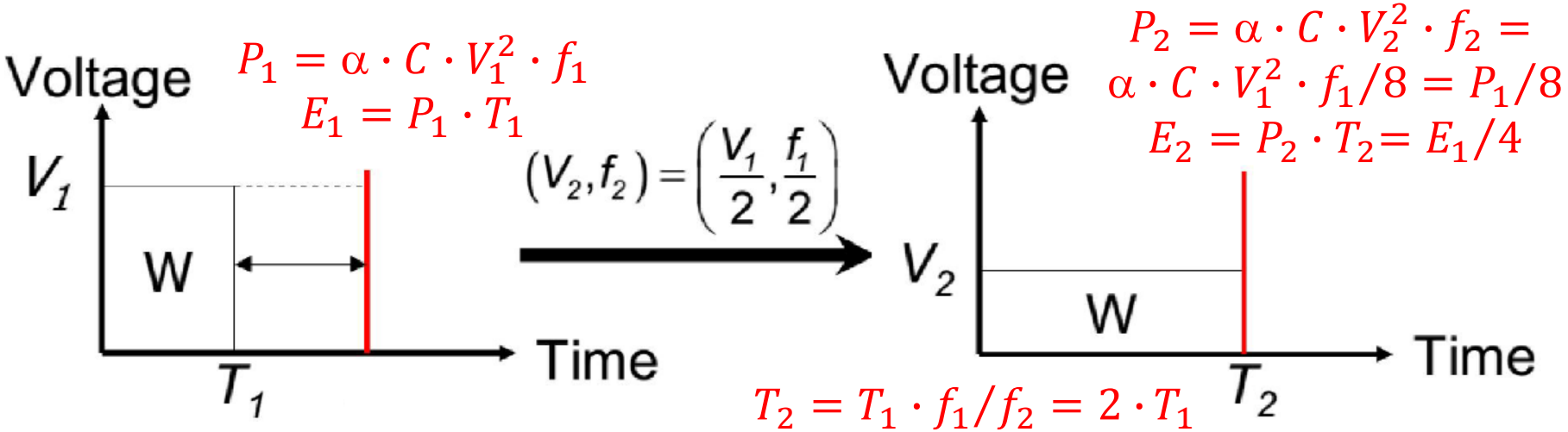
$$P \sim \alpha C_L V_{dd}^2 f$$

V_{dd}: supply voltage C_L : load capacity
α : switching activity f : clock frequency

- IDEA: keep the host active but lower the CPU voltage/frequency to save power

Active low-power modes

- Power saving comes at a performance cost as there is an increase in the execution time
  - Example: reducing voltage and frequency by half, divides power by 8, but if $T_2 = 2 \cdot T_1$, static energy doubles and dynamic energy is only reduced by 4

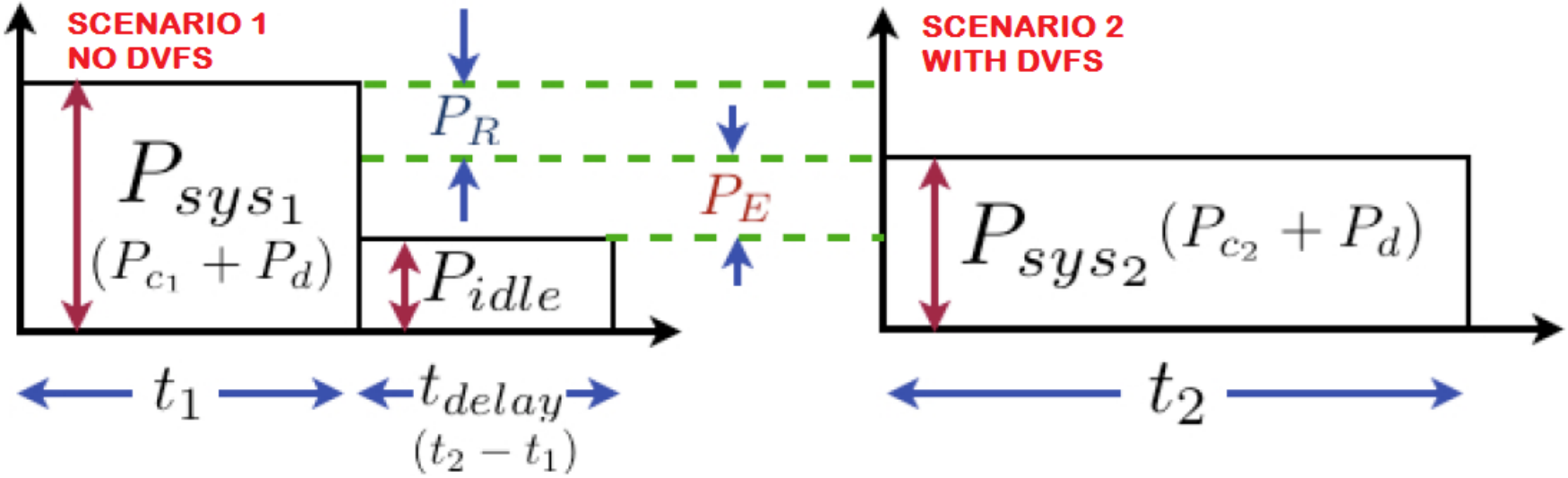


Courtesy of [ChPe05]

Active low-power modes

- Only the time spent in the CPU and the CPU power are sensitive to DVFS
  - $t_2 = t_{c_2}(\uparrow) + \boldsymbol{t_d}$
  - $P_{sys_2} = P_{c_2}(\downarrow) + \boldsymbol{P_d}$

t_d: time in other resources
P_d: power of other resources
t_c: CPU time
P_c: CPU power



Active low-power modes

- DVFS only provides energy savings depending on the relation between the execution time increase and the CPU power reduction

$$P_{sys_1} \cdot t_1 + P_{idle} \cdot t_{delay} > P_{sys_2} \cdot (t_1 + t_{delay})$$

$$(P_{sys_1} - P_{sys_2}) \cdot t_1 > (P_{sys_2} - P_{idle}) \cdot t_{delay}$$

$$(P_{c_1} + P_d - P_{c_2} - P_d) \cdot t_1 > (P_{c_2} + P_d - P_{c_{idle}} - P_{d_{idle}}) \cdot t_{delay}$$

$$\underbrace{(P_{c_1} - P_{c_2})}_{} \cdot t_1 > \underbrace{(P_{c_2} - P_{c_{idle}} + P_d - P_{d_{idle}})}_{} \cdot t_{delay}$$

$\boldsymbol{P_R}$: reduction in CPU power consumption due to DVFS

$\boldsymbol{P_E}$: extra power consumption due to DVFS

- CPU power spent instead of being idle
- Power spent by other resources

Contents

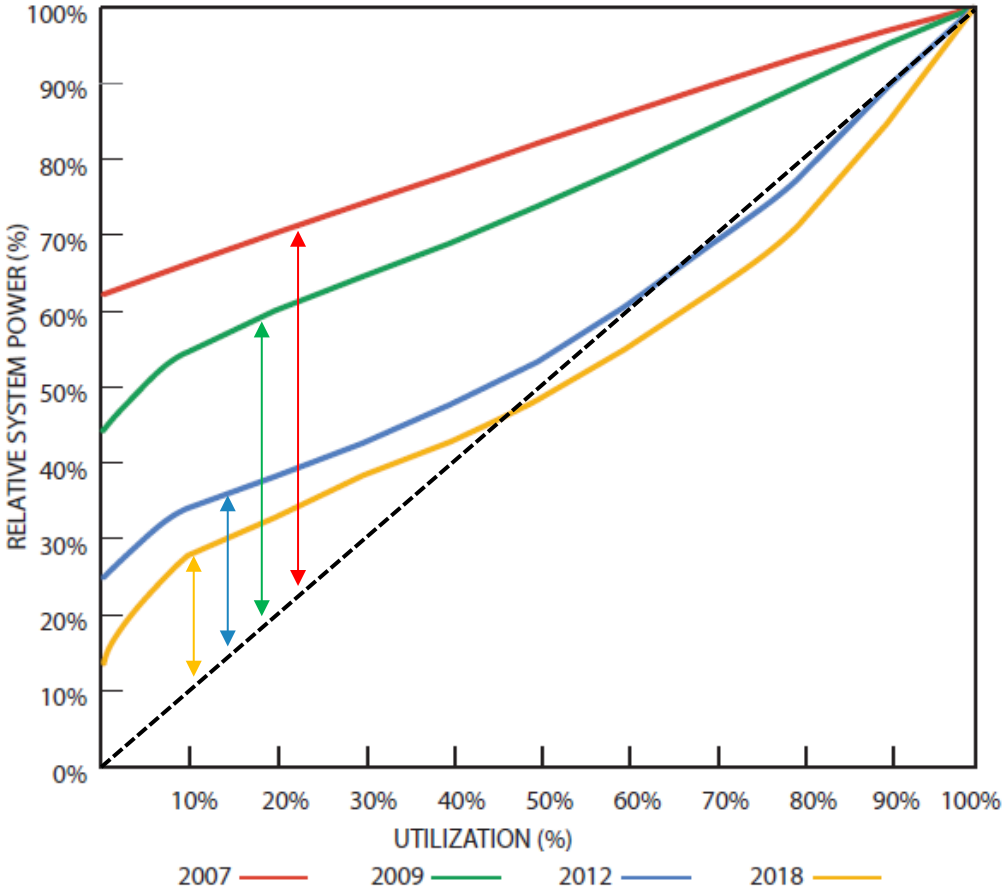
- Energy awareness
- **Efficiency of the IT equipment**
  - Low-power modes
  - **Resource allocation and scheduling**
- Cooling subsystem
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Resource allocation and scheduling

- Adapt policies to optimize energy efficiency
  - When to run each task
  - How many PEs to assign to each task
  - Which PEs to assign to each task
  - What power state to set for each PE
- Can take profit of the resource management capabilities of **virtualization** technology
  - VM consolidation
  - Live VM migration
  - VM allocation

Energy proportionality

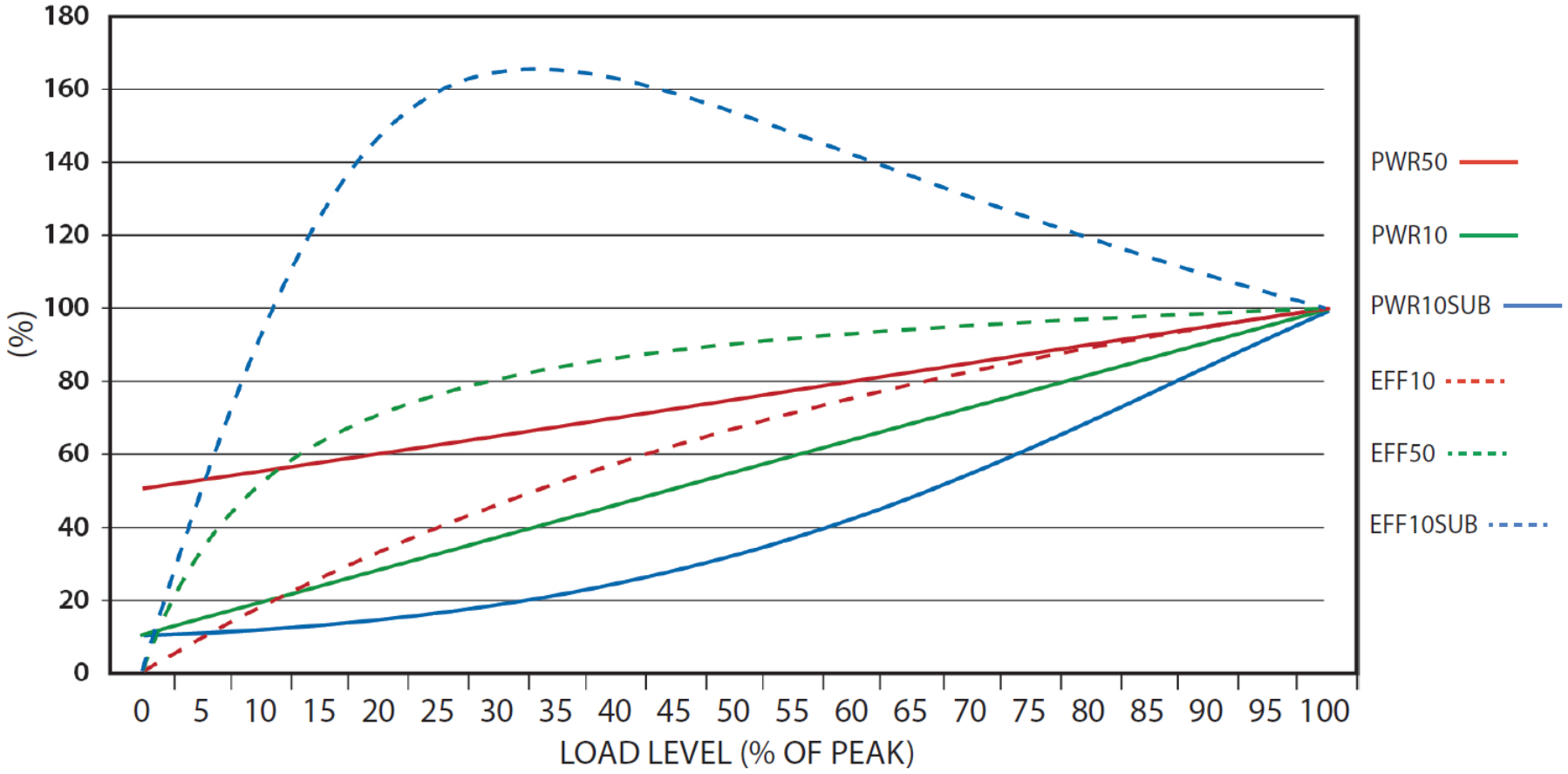
System power is not proportional to the load (esp. at lower loads)



Courtesy of [BaHR19]

Energy proportionality

Non-proportional power reduces energy efficiency at lower loads



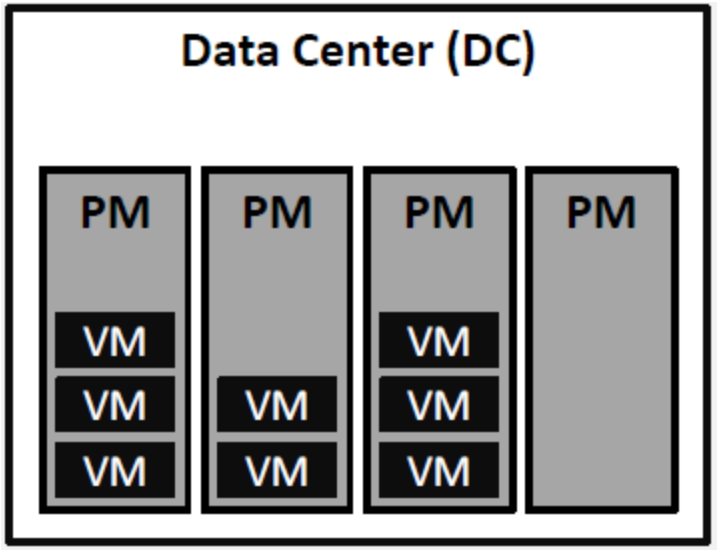
Courtesy of [BaHR19]

VM consolidation

- We can play with VM consolidation to:

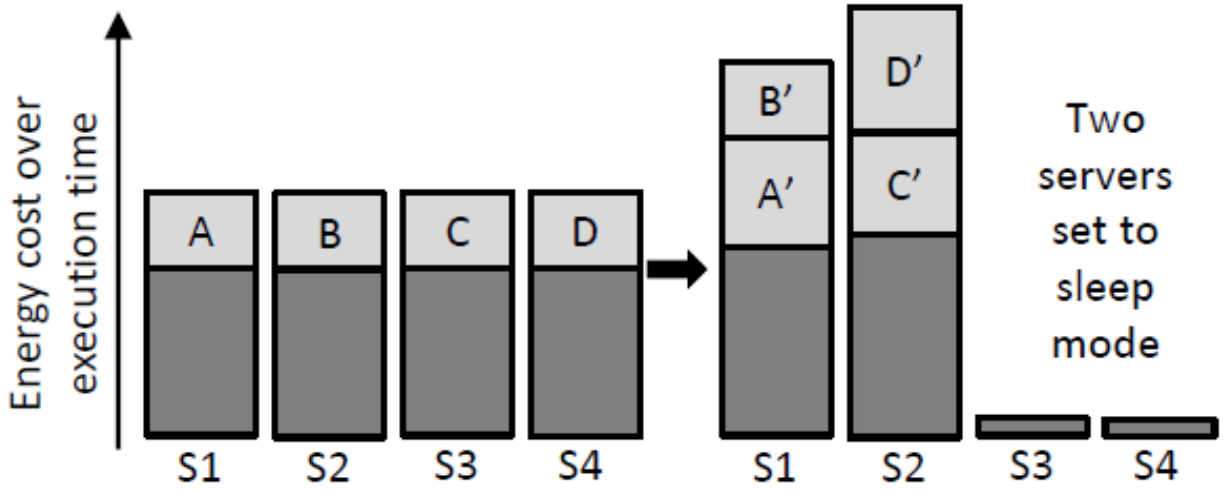
1. Run the hosts at higher utilizations closer to its most energy-efficient operating point
   - Live VM migration supports dynamic consolidation
2. Reduce the number of hosts needed to serve a given workload

   ⇒ Use of inactive low-power modes



VM consolidation

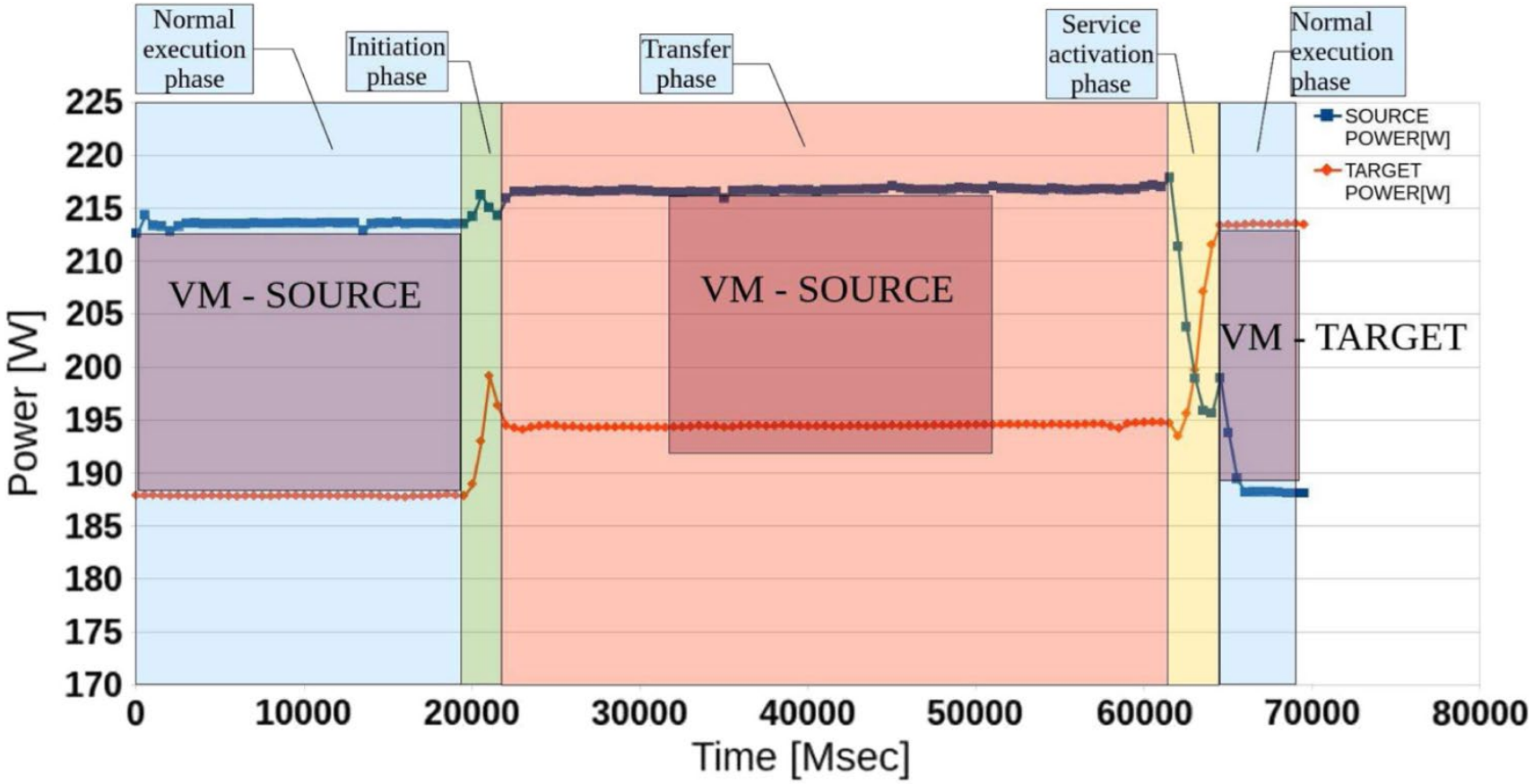
- VMs co-located in the same physical host may suffer resource contention (a.k.a. interference)
  - Run time will increase, thus also static and dynamic energy, wiping out some savings obtained by turning off servers
- Contention-aware scheduling considers the impact of co-locating VMs when deciding their placement



Courtesy of [RKGL13]

Live VM migration

- It consumes energy! $\Rightarrow E_{migr} = E_{init} + E_{transfer} + E_{activ}$



Courtesy of [MPBK16]

Live VM migration

- Most impacting parameters on E_{migr}
  1) CPU utilization of the source and target hosts
     a) CPU utilization for each VM executing in the host
        – During the initiation and transfer, the source includes the CPU utilization of the migrating VM
     b) CPU utilization of the hypervisor
     c) CPU utilization added by the migration (i.e., VM suspension/restart, state transfer, …)
  2) Memory allocated to the migrating VM
  3) Memory dirtying rate of the migrating VM
  4) Network bandwidth between the source and target for transferring the state of the VM

E_{init} & E_{activ} depend on 1)

$E_{transfer}$ depends on 1), 2), 3), and 4)

VM allocation

- Recall that VM allocation can be viewed as a packing problem, similar to bin-packing
- Energy-aware VM allocation must consider more challenges than bin-packing (apart from those discussed in Lesson 3)
  1. The power consumption of a host regarding the load is non-proportional
  2. Hosts can have heterogeneous power consumption profiles regarding the load
  3. VM migrations incur an energy cost

WSCs ecosystems

- WSCs can work within ecosystems where they interact with other WSCs
  - e.g. lease capacity to meet demand peaks
- IDEA: we can select the most appropriate WSC where to deploy every workload
  - e.g. follow-the-moon strategy: deploy workloads by following the night-time
    - Easier workload allocation, as target WSCs are in an off-peak demand interval
    - Reduced costs, as electricity is generally cheaper at night
    - The lower outside temperature favors using free cooling

WSCs ecosystems

- There are new opportunities for energy-driven management, but also new challenges
  a) The distant geographic locations of WSCs
     - Increased performance cost (and consumed energy) for the migration of VMs
     - Increased communication latency in the data exchange among VMs located in different WSCs
  b) The independent administrative domains involved in the ecosystem
     - Conflicting goals among the WSCs
     - Lack of information about the energy consumption and energy mix of the other WSCs

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Cooling subsystem

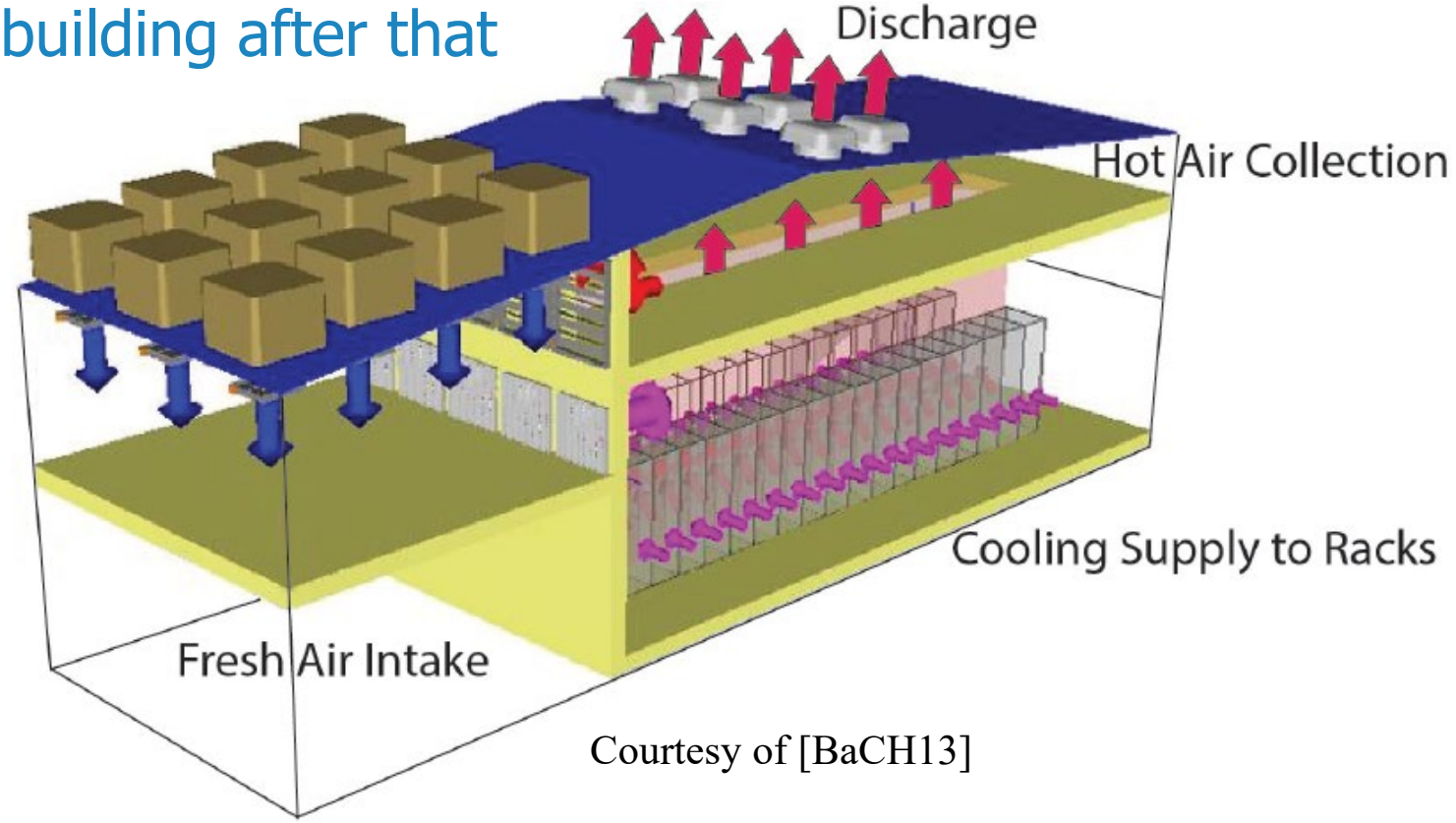
- Remove heat generated by all the equipment
- Using some hierarchy of loop systems
  - Each loop brings in a cold medium that warms up via some form of heat exchange and is somehow cooled back again
- Open loop: replace the outgoing warm medium with a cool supply from the outside
- Closed loop: recirculate the same medium again and again, transferring heat to an adjacent upper loop in a heat exchanger

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- Energy awareness
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- **Cooling subsystem**
  - **Free cooling**
  - CRAC-based cooling
  - High-density cooling
  - Efficiency of cooling subsystem
- Power supply subsystem

Free cooling

- Open single loop system that cools the servers by flowing outside air, which is expelled outside the building after that



Courtesy of [BaCH13]

Free cooling

- Advantages
  - ↑ The most efficient cooling system
- Disadvantages
  - ↓ Does not work in all climates
  - ↓ May require auxiliary cooling when external conditions are not favorable
  - ↓ Requires filtering to protect from air contamination
  - ↓ Temperature may be difficult to control
    - If there is a malfunction, temperature rises very quickly

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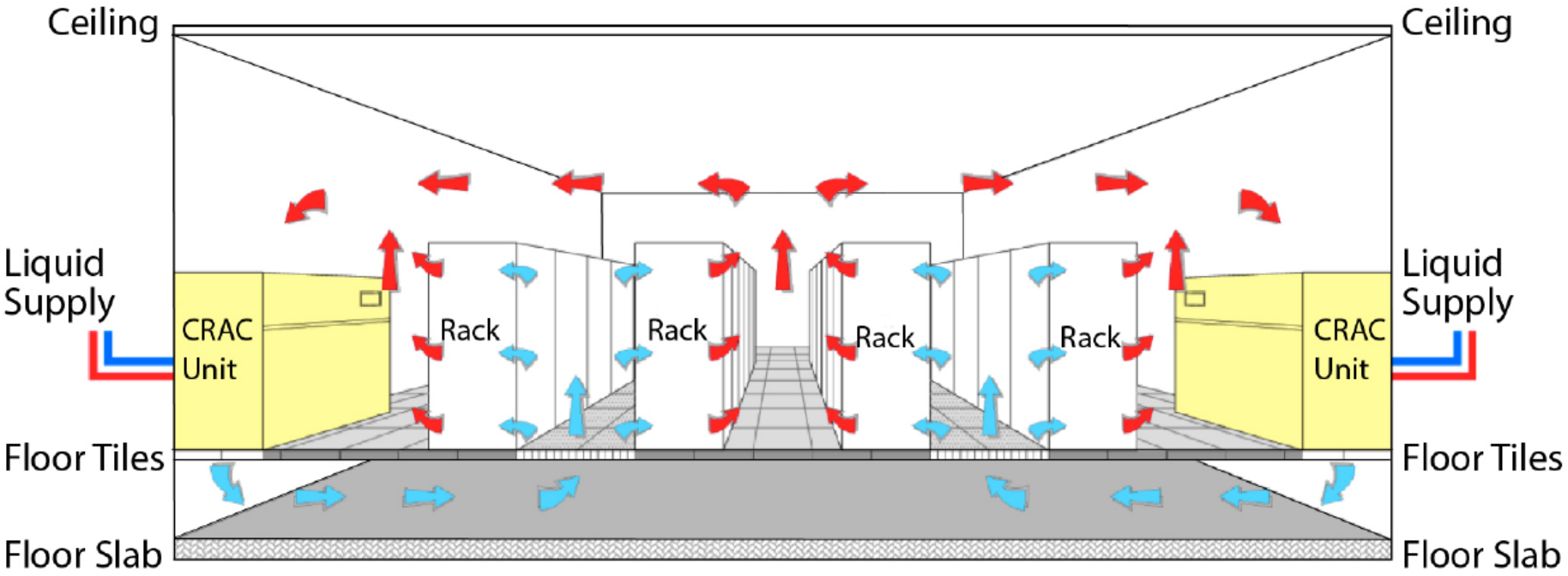
CRAC-based cooling

- Closed-loop system with two loops:

1. **Air circuit** where cold air flows from CRACs to servers, heats up, and eventually reaches a heat exchanger to cool it down again for the next cycle through the servers
2. **Liquid supply** which leads directly from the CRACs to external heat exchangers that discharge the heat to the environment

CRAC: Computer Room Air Conditioning

CRAC-based cooling



Courtesy of [BaCH13]

↑ Easy to implement

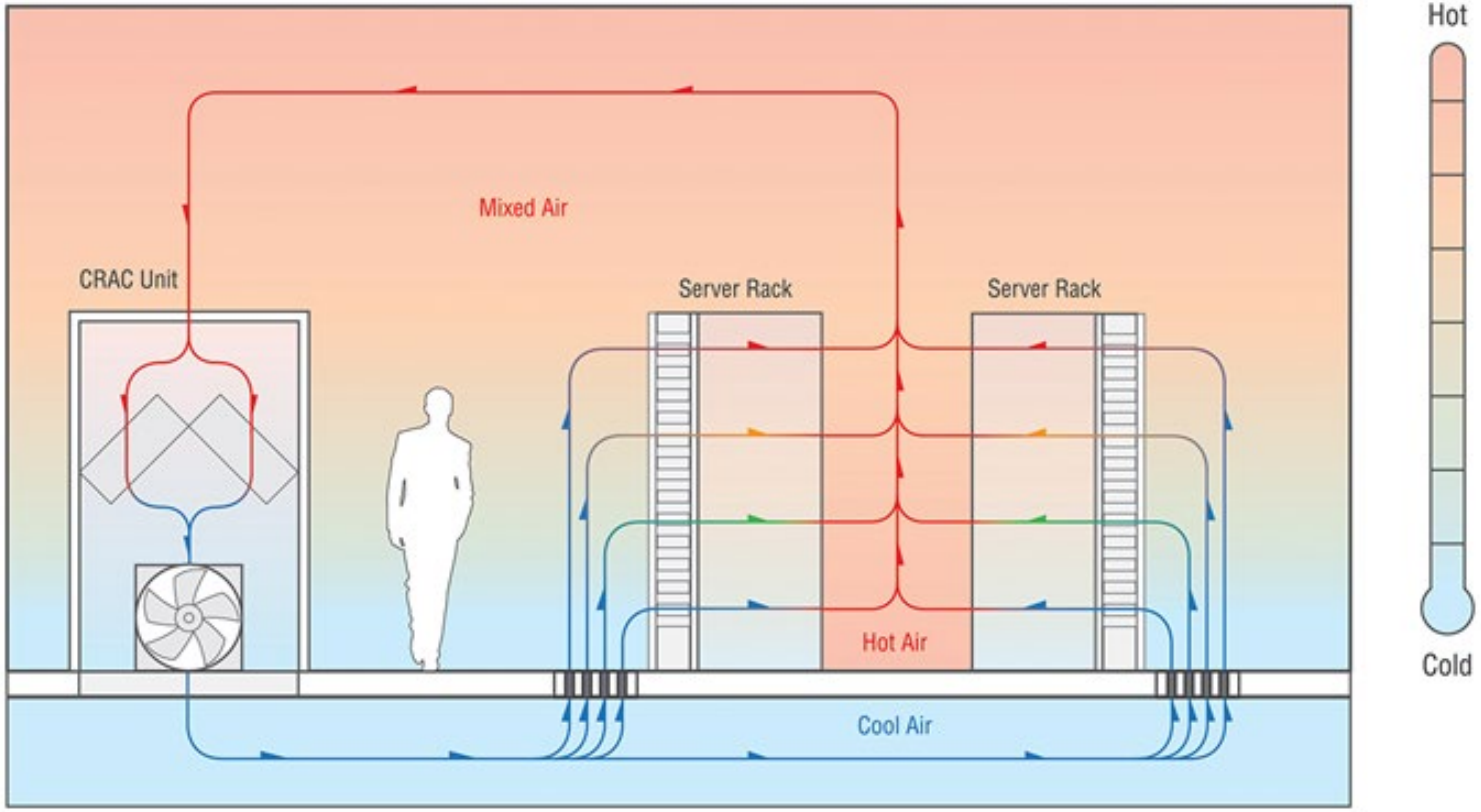
↑ Relatively inexpensive to construct

↑ Protection from external contamination

↓ Low operational efficiency

CRAC-based cooling

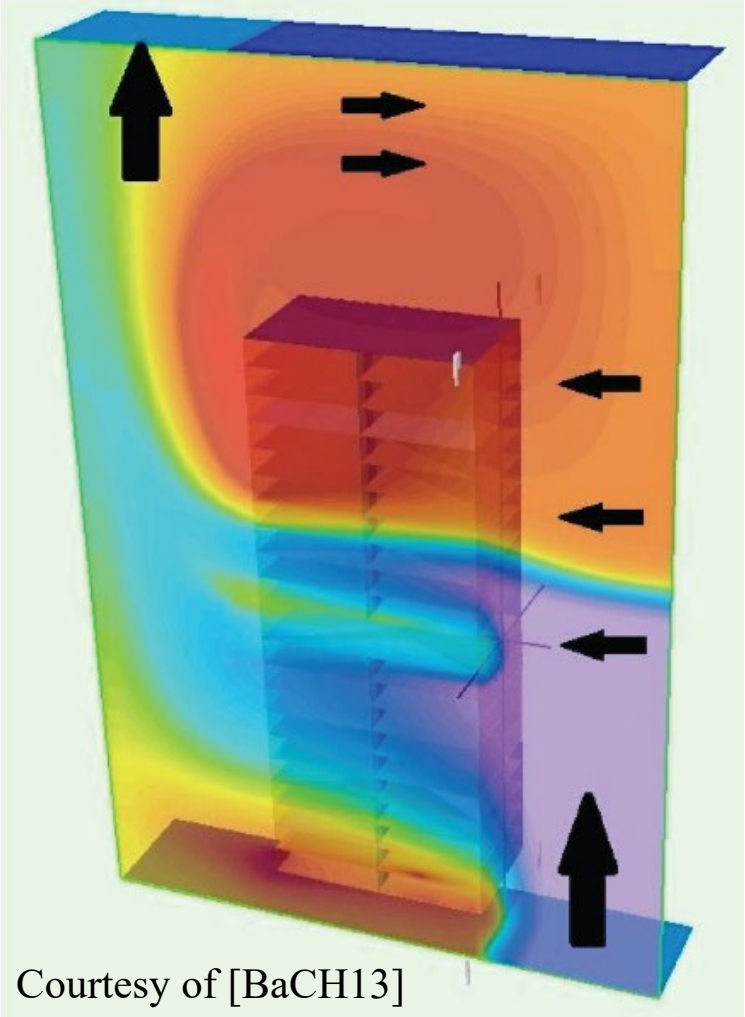
- Insufficient air flow is what limits efficiency because hot air can recirculate



Courtesy of [Sass15]

CRAC-based cooling

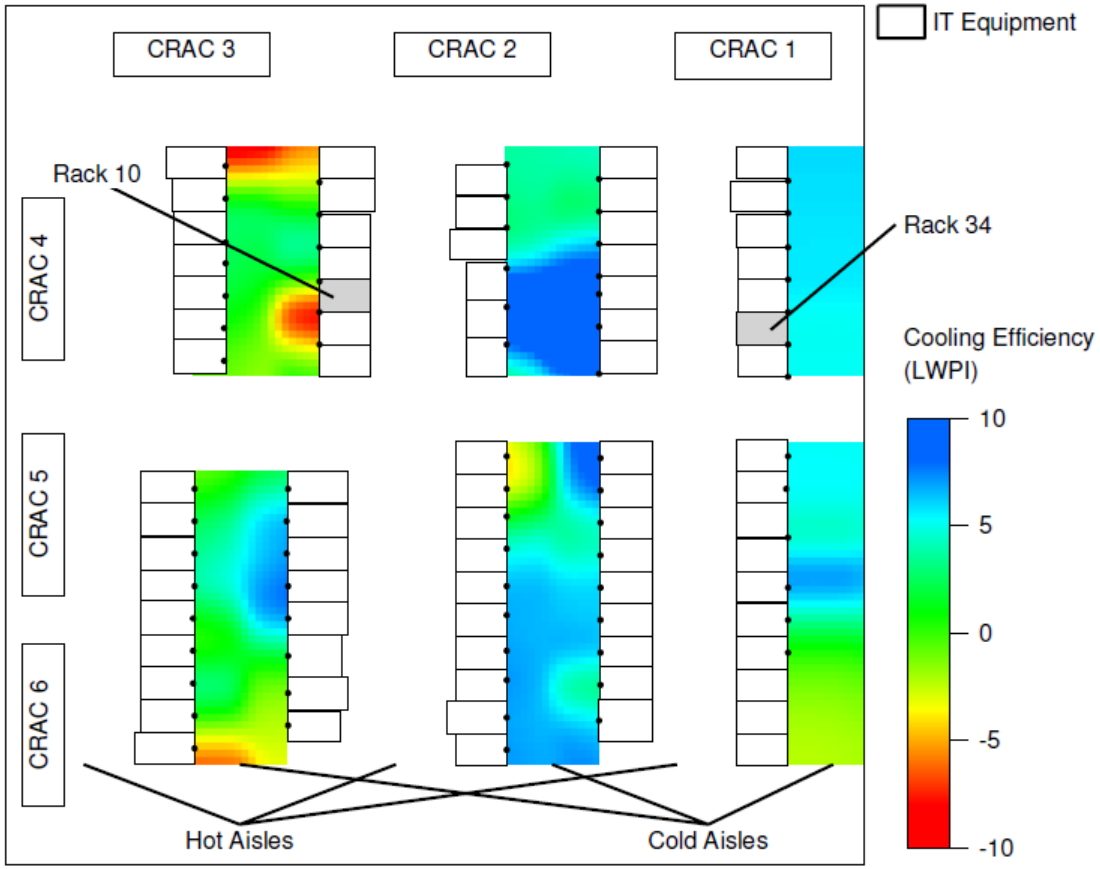
- When the cold air flow coming from the floor does not match the horizontal air flow through the servers, some of them will ingest recirculated hot air
- The typical response is to lower the temperature of the CRAC output
- That works but increases energy costs significantly



Courtesy of [BaCH13]

CRAC-based cooling

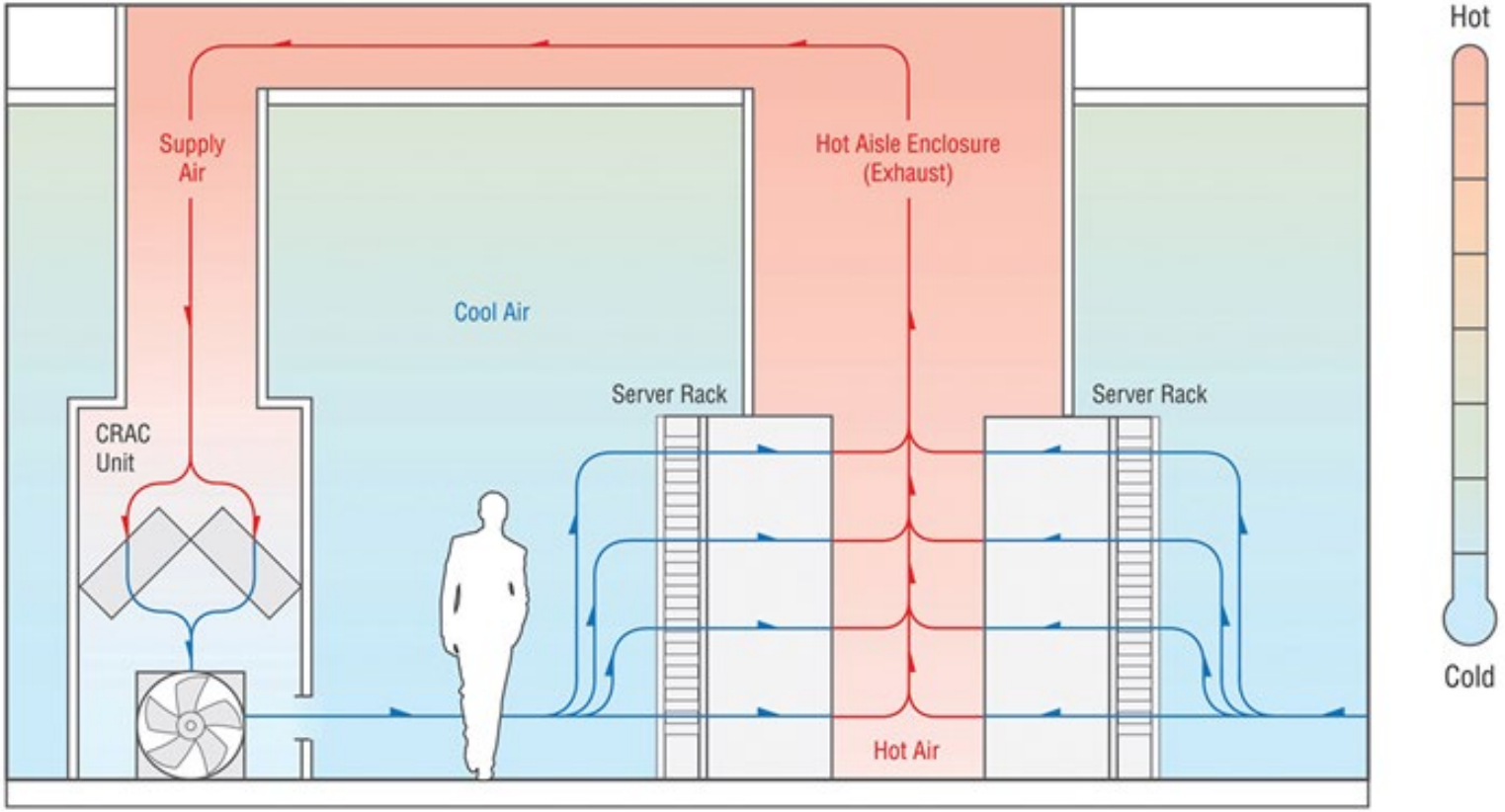
- Insufficient air flow also makes some locations in a WSC to be more efficiently cooled than others



Courtesy of [CGHW10]

CRAC-based cooling

- Air flow problems can be attenuated by physically separating cold and hot air as much as possible



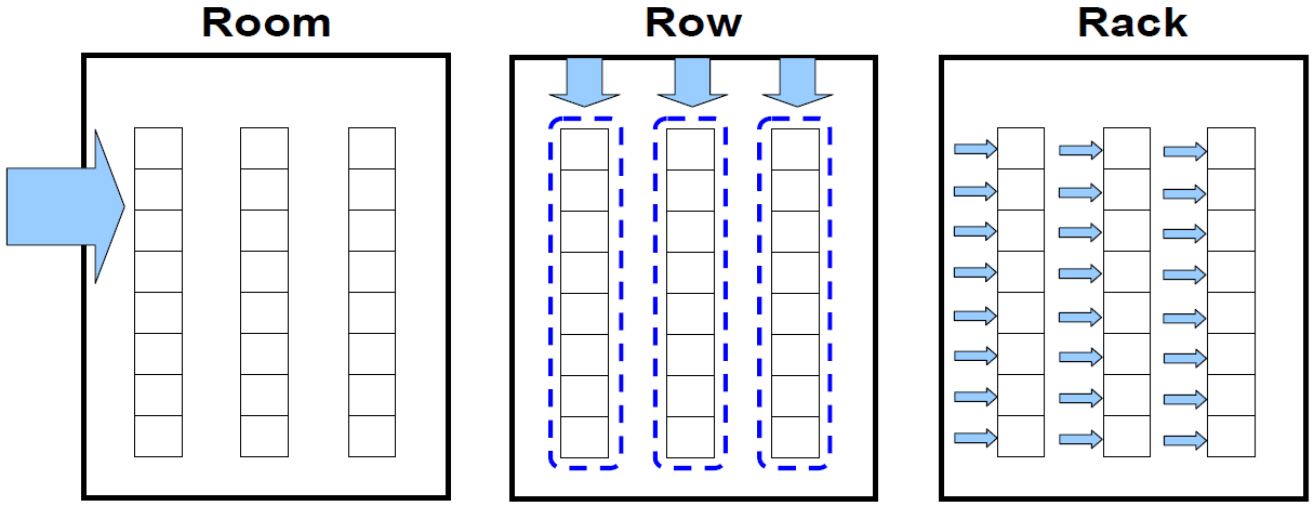
Courtesy of [Sass15]

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High-density cooling

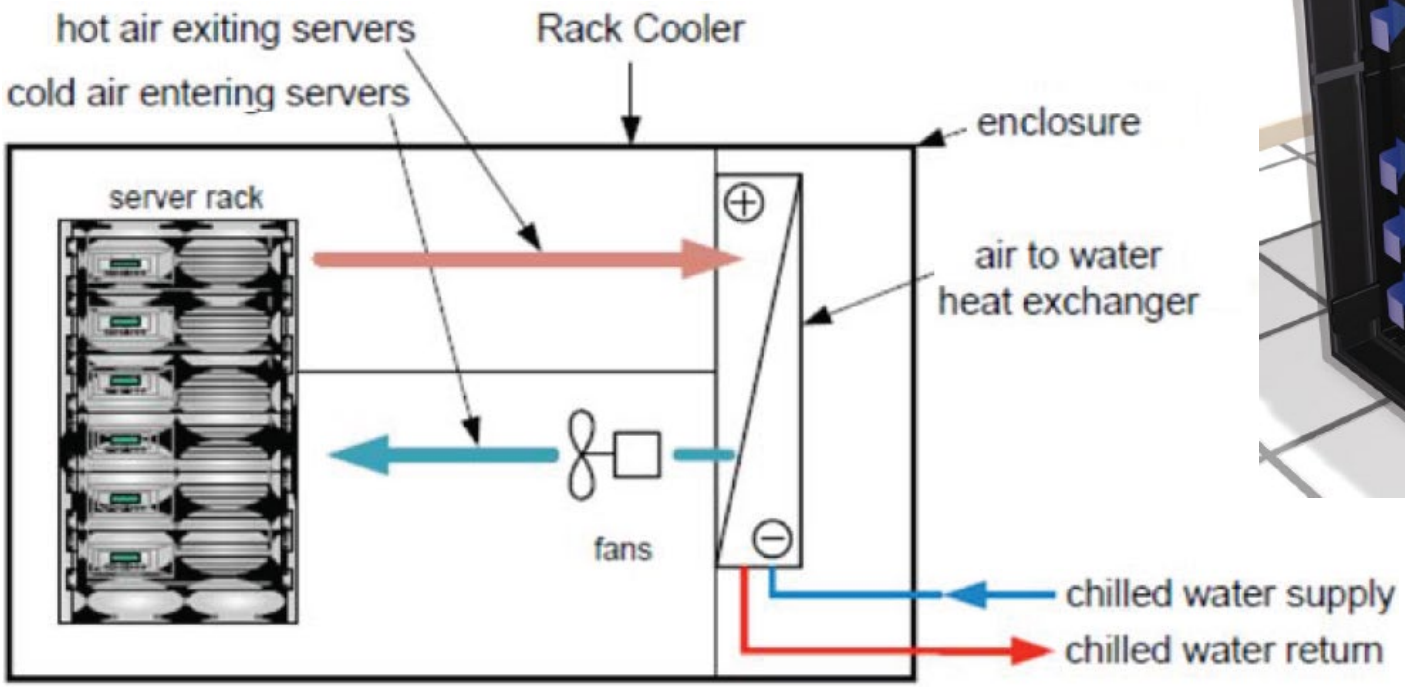
- Physical limits to generate air flow from the raised floor restrict the power density of WSCs
- We need other techniques to increase power density and cooling efficiency
- ➢ Associate CRAC units with rows of racks or individual racks instead of the whole room



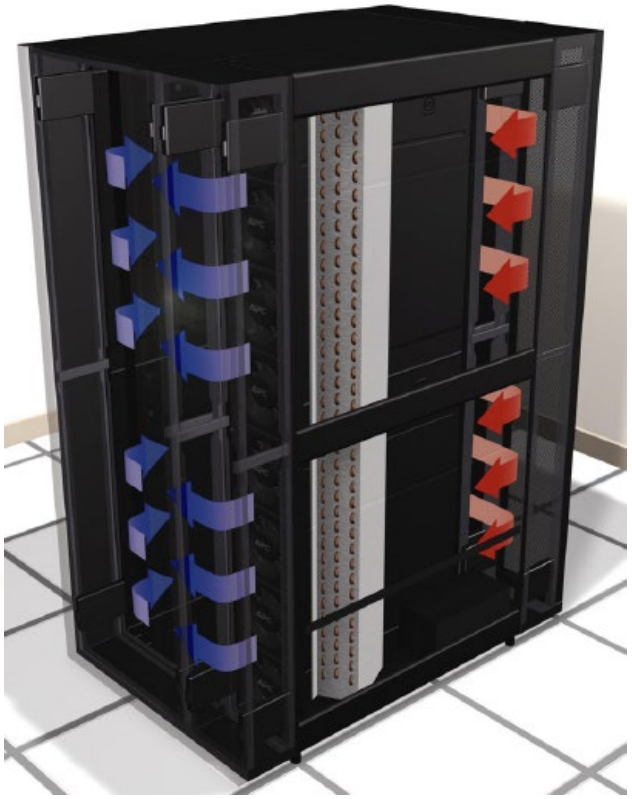
Courtesy of [DuRa06]

In-rack cooling

- Add an air-to-water heat exchanger at the back of a rack
  - Hot air exiting the servers immediately flows over coils cooled by water



Courtesy of [Bell12]

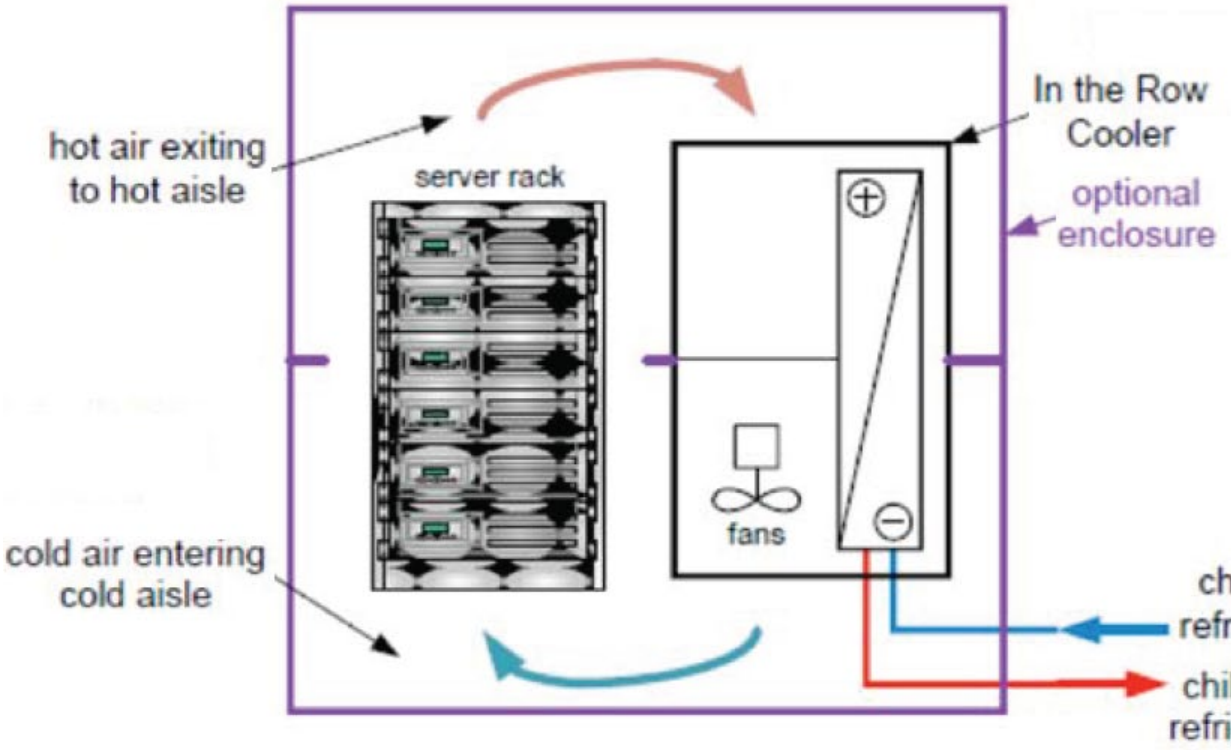


Courtesy of [DuRa06]

In-row cooling

- Cooling units sit within the rows of racks

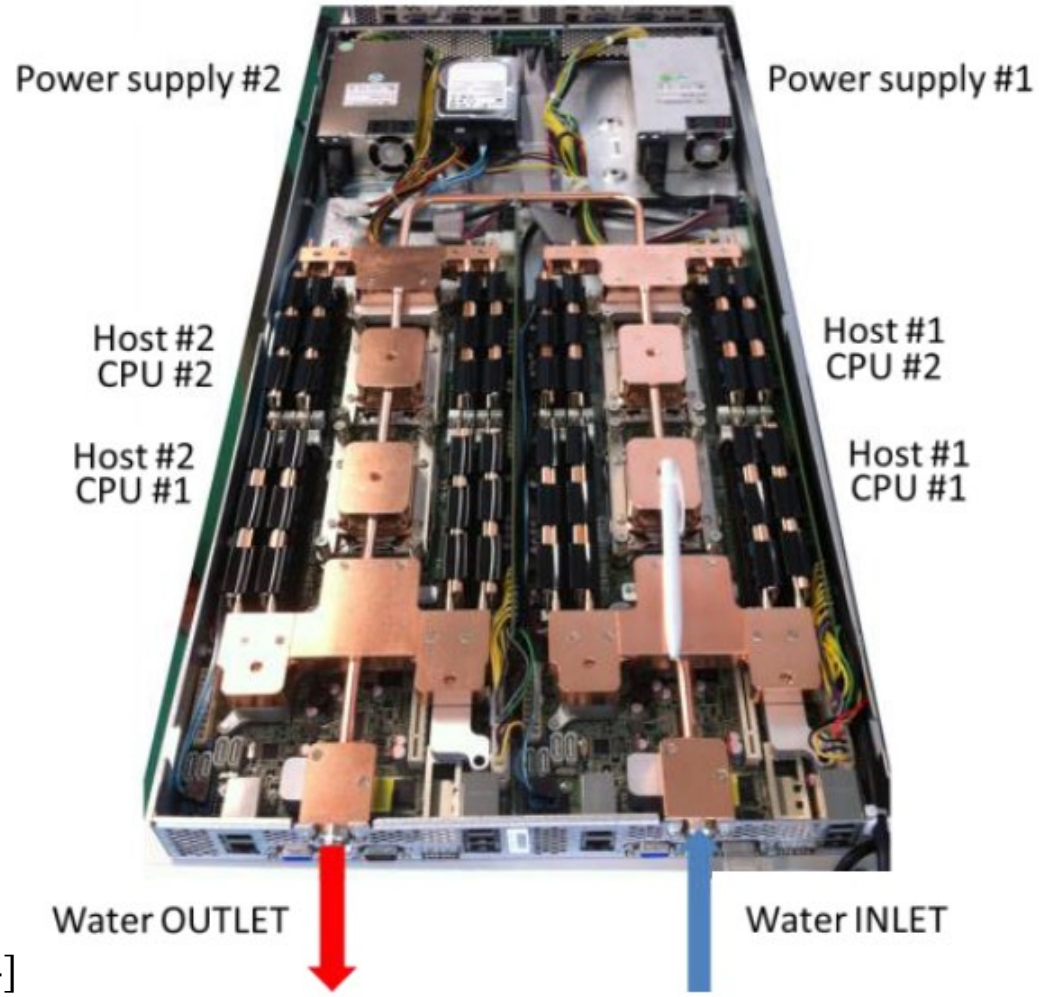
Courtesy of [Bell12]





Cold plates

- Directly cool server components using local, liquid-cooled heat sinks



Courtesy of [OPMN14]

High-density cooling

- Advantages
  - ↑ Higher power density
  - ↑ Better cooling efficiency
    - Due to the reduction of the hot air path between server exhaust and CRAC input
- Disadvantages
  - ↓ Chilled water needs to be brought to each rack/row/plate, greatly increasing the cost and the complexity of design
  - ↓ Danger of water leakage

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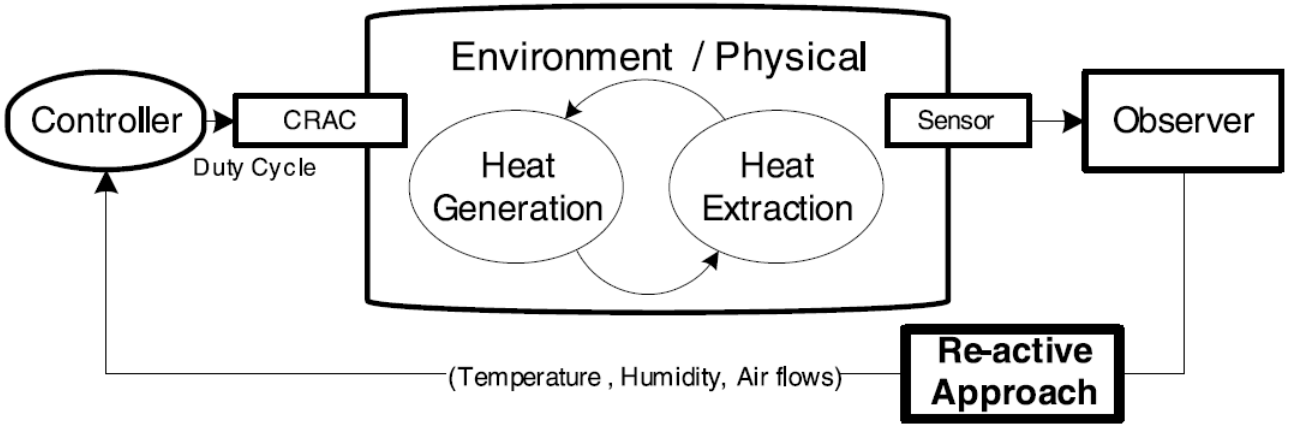
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  - High-density cooling
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- Power supply subsystem

Efficiency of cooling subsystem

- We discussed how to reduce the consumption of these systems from a HW perspective
  a) Improve air flow handling by segregating hot air exhausted by servers from cold air and keeping the path to the cooling unit short
  b) Elevate the temperature of the cold aisle, which makes it much easier to cool the WSC efficiently
     – IT equipment can work correctly at higher temperatures than commonly used
  c) Use free cooling, especially in moderate climates

Efficiency of cooling subsystem

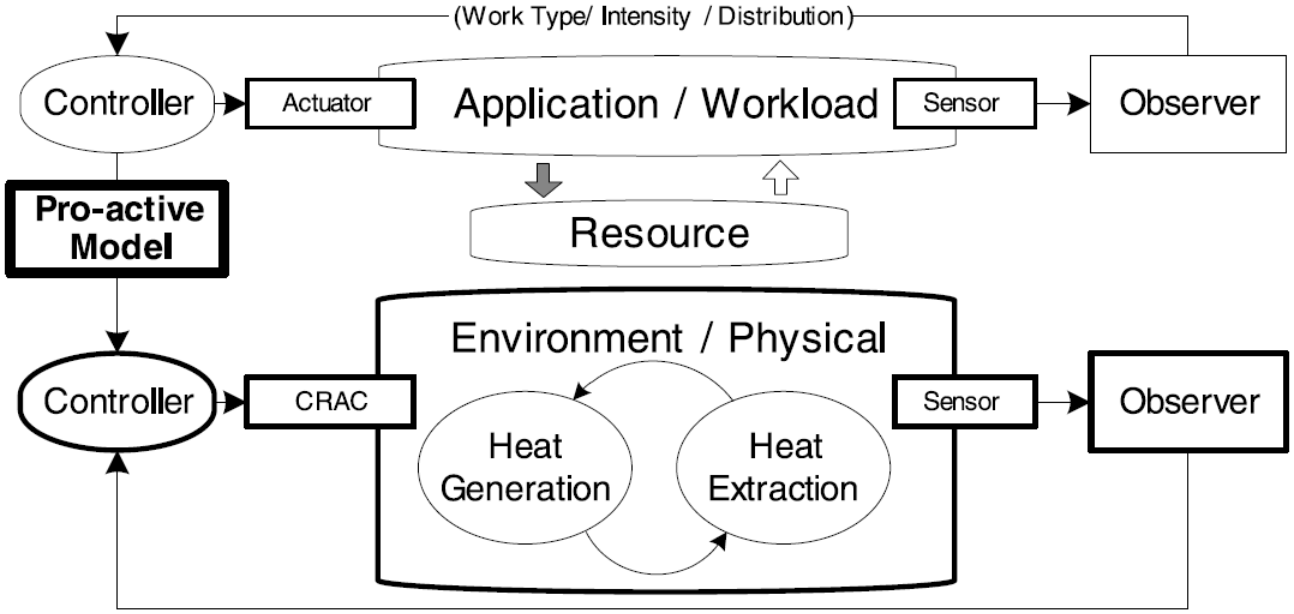
- Typically, the cooling operation is triggered when temperature surpasses predefined thresholds



- Further improvements are possible if cooling is managed together with the IT equipment

Efficiency of cooling subsystem

a) Adapt <u>proactively</u> the amount of cooling to the expected workload
   - e.g. throttle down the cooling when low IT workload is expected in a near future



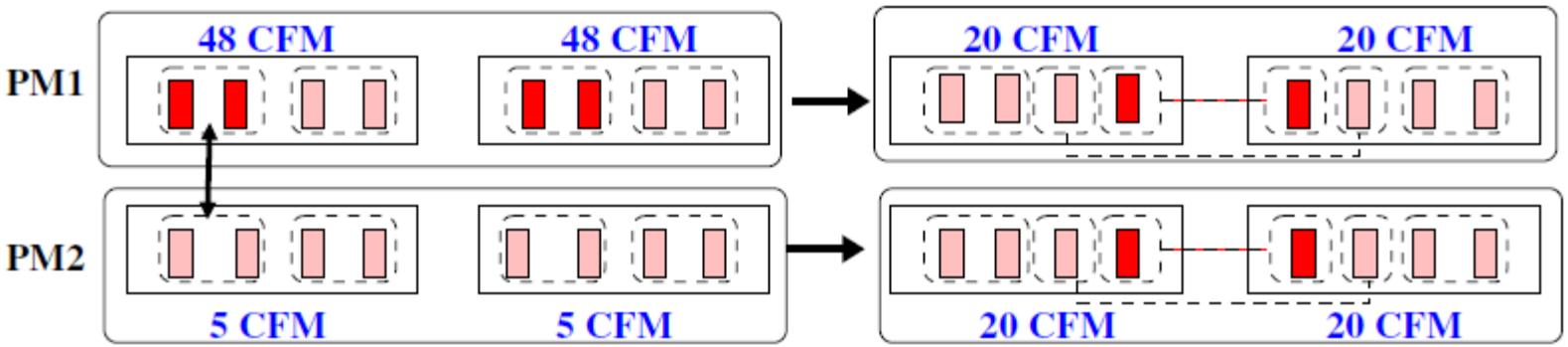
Courtesy of [LKPP12]

Efficiency of cooling subsystem

b) Consider the resulting temperature of the hosts as a criterion to decide the allocations

1. Keep power density (and dissipated heat) of the servers in check to prevent 'hot spots'
   - Spread the tasks to balance the load of the servers
   - Delay task start or extend task execution time (allocating fewer resources or through DVFS) to smooth the load

CFM (cubic feet per minute): airflow measurement

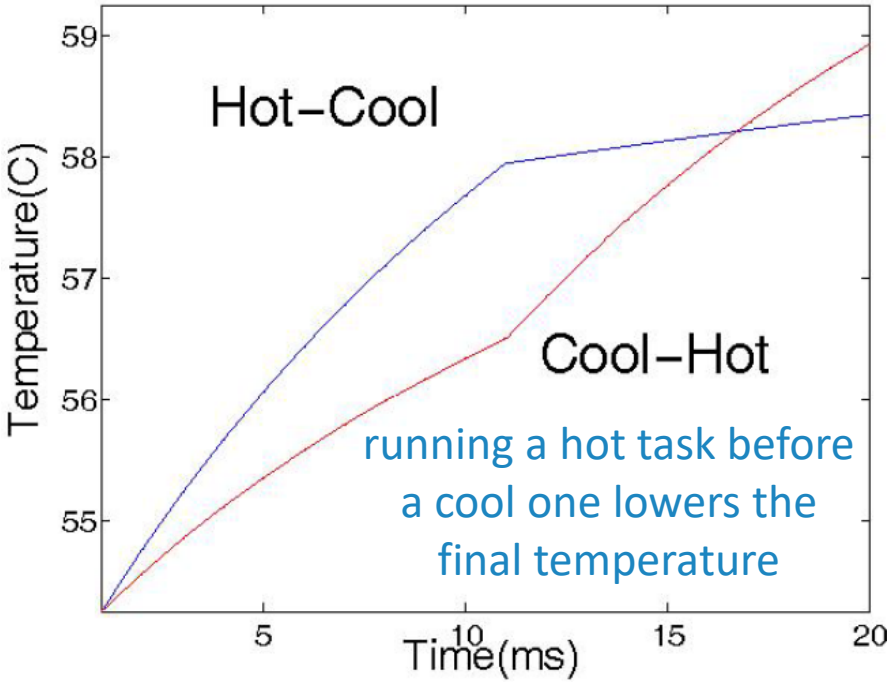


Courtesy of [AySR10]

Efficiency of cooling subsystem

b) Consider the resulting temperature of the hosts as a criterion to decide the allocations

2. Place hot tasks in the hosts of the WSC that are easier to cool (refer to slide 51)
3. Schedule according to the impact of tasks in the temperature
   - Run first the hottest task that will not increase the temp. above a threshold



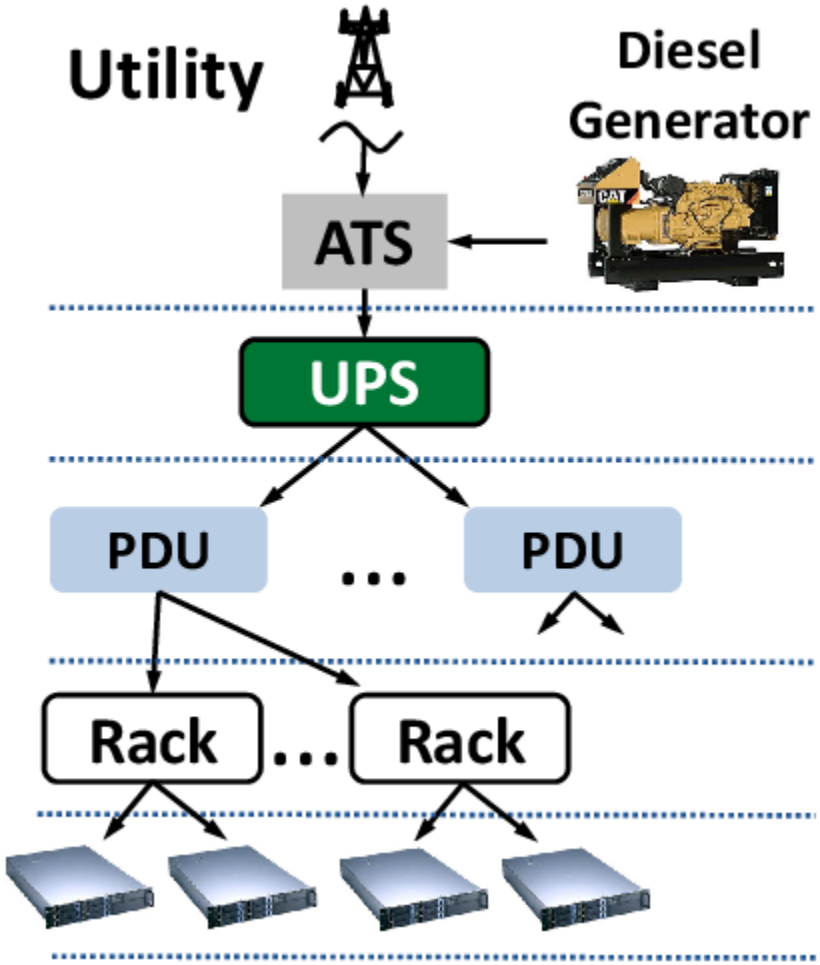
Courtesy of [YZCZ08]

Contents

- Energy awareness
- Efficiency of the IT equipment
- Cooling subsystem
- **Power supply subsystem**

Power supply subsystem

- Power enters the WSC going to the UPS system
- UPS also takes another feed from a diesel generator that bursts in when utility power fails
- Output lines from UPS are routed to PDUs in the WSC floor
- PDUs route individual power circuits to racks

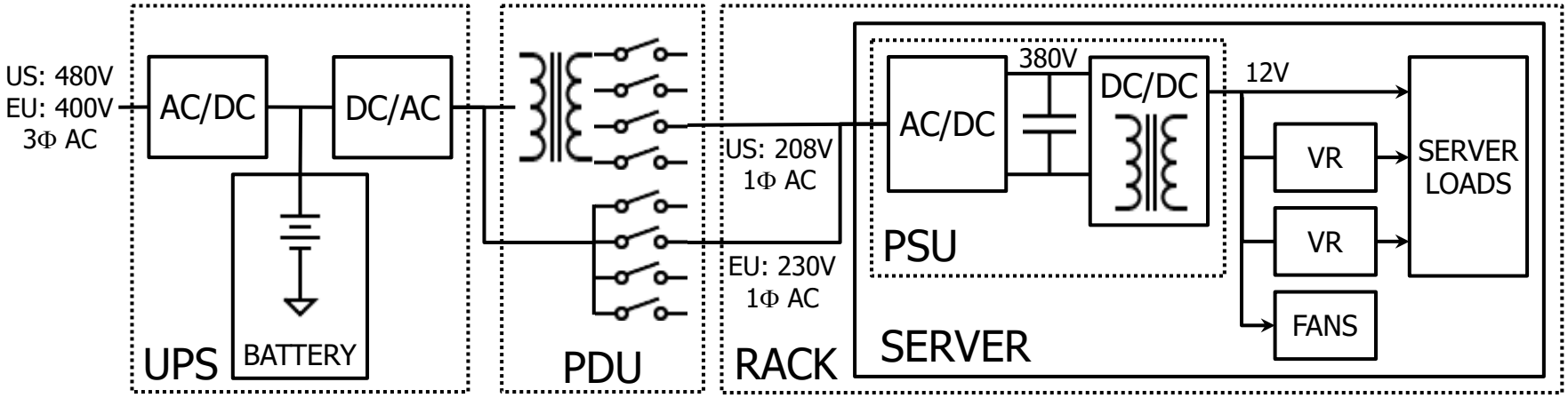


Uninterruptible Power Supply (UPS)

1. Chooses the active power input
   a) Utility grid power
   b) On-site power
      - Diesel generator (used when the utility power fails)
      - Renewable sources (if available)
2. Contains some form of energy storage
   – To bridge the time between the utility failure and the availability of on-site power
3. Conditions incoming power feed, removing voltage spikes (through double conversion)
   – Some power efficiency is lost due to conversions

Uninterruptible Power Supply (UPS)

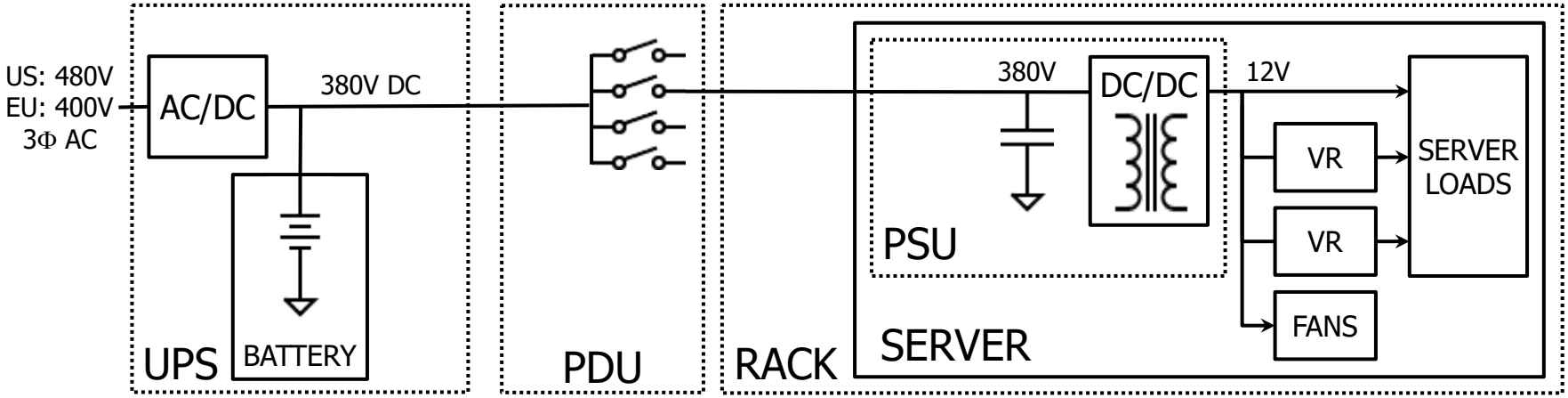
- Traditional AC–DC–AC double conversion
  - Input AC is rectified to DC, which feeds batteries
  - DC is then inverted back to AC to feed the PDUs
  - If utility AC power fails, DC remains from batteries
    - Eventually, generator starts and resupplies AC power
  - State-of-the-art efficiency around 85%



Courtesy of [BaHR19]

Uninterruptible Power Supply (UPS)

- DC systems eliminate the final inversion step of the UPS (DC to AC)
  - They can also eliminate PDU power transformation and PSU AC/DC rectification
  - State-of-the-art efficiency around 89%



Courtesy of [BaHR19]

Uninterruptible Power Supply (UPS)

- Advantages of DC systems vs. AC systems
  - ↑ (Slightly) Higher power efficiency
  - ↑ Higher reliability from reduced component counts
  - ↑ Easier integration of distributed generators with native DC output
    - e.g. solar photovoltaic, fuel cells, and wind turbines
- Disadvantages
  - ↓ Costs of commercial DC equipment are currently higher than comparable AC equipment
  - ↓ DC technology for WSCs is less mature
  - ↓ Limited availability of DC technicians

Power Distribution Unit (PDU)

- They take a larger input feed and break it up into many smaller circuits that distribute power to the actual servers
- Each circuit is protected by its own breaker so a short in a server or power supply will trip only the breaker for that circuit, not the entire PDU or even the UPS
  - PDUs resemble breaker panels in our houses

$\Rightarrow$ We can reduce UPS and PDU losses by selecting higher-efficiency gear

Contents

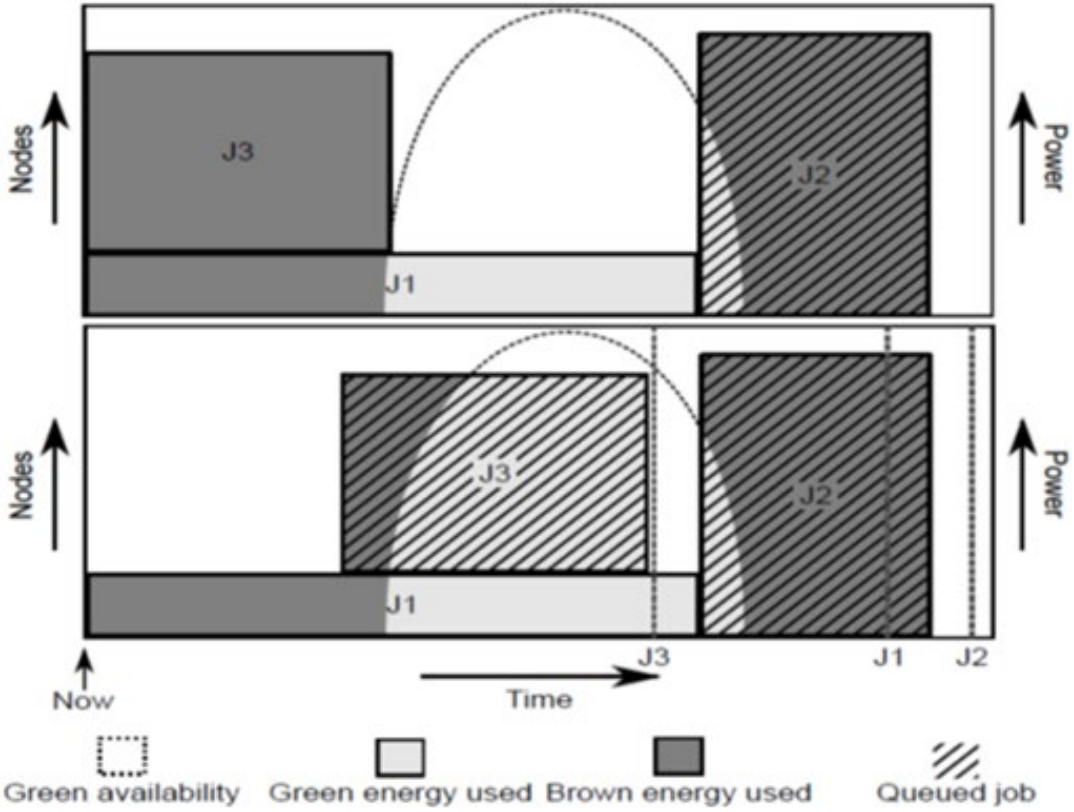
- Energy awareness
- Efficiency of the IT equipment
- Cooling subsystem
- **Power supply subsystem**
  - **Renewable energy**
  - Efficiency of power supply subsystem

Renewable energy

- Sustainable WSCs require using as much as possible energy sources that do not emit CO_2
- Using intermittent energy sources (e.g. solar and wind energy) is challenging, as energy is not always available
- Adapt the IT workload to the available energy (i.e. workload shaping)
  1. Use DVFS or server power state tuning to adjust the IT power demand to the time-varying renewable power budget

Renewable energy

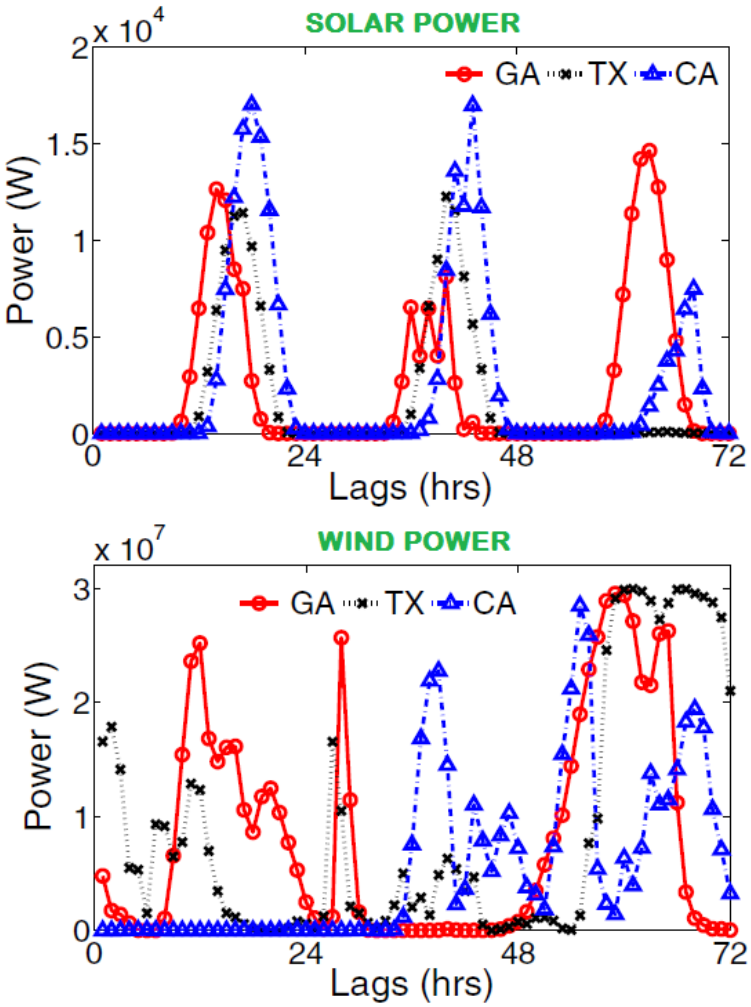
2. Use task scheduling to postpone tasks when not enough renewable energy is available to execute them



Courtesy of [GHLB15]

Renewable energy

- Ecosystems offer more opportunities to increase the use of renewable energy, as they comprise WSCs located in different geographical locations
  - e.g. follow-the-renewables strategy: deploy workloads by following the availability of renewable energy in the various WSCs



Courtesy of [AbPG14]

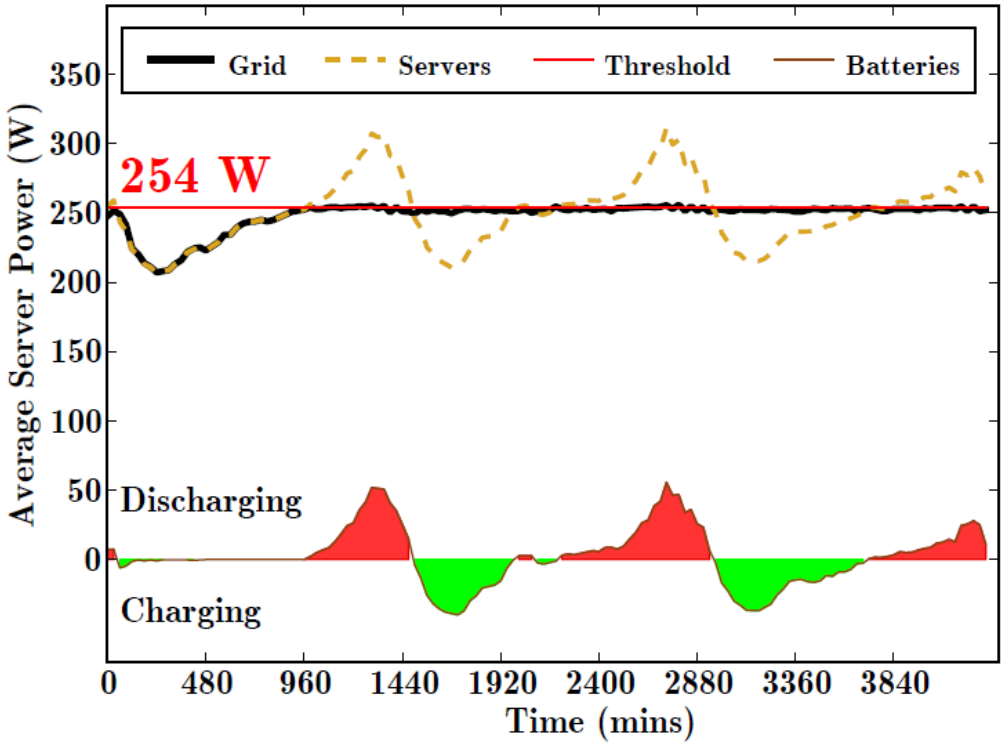
Contents

- Energy awareness
- Efficiency of the IT equipment
- Cooling subsystem
- **Power supply subsystem**
  - Renewable energy
  - **Efficiency of power supply subsystem**

Usage of energy storage

A. Manage short demand peaks or short-term (renewable) supply reductions
   - Use stored energy instead of capping the load

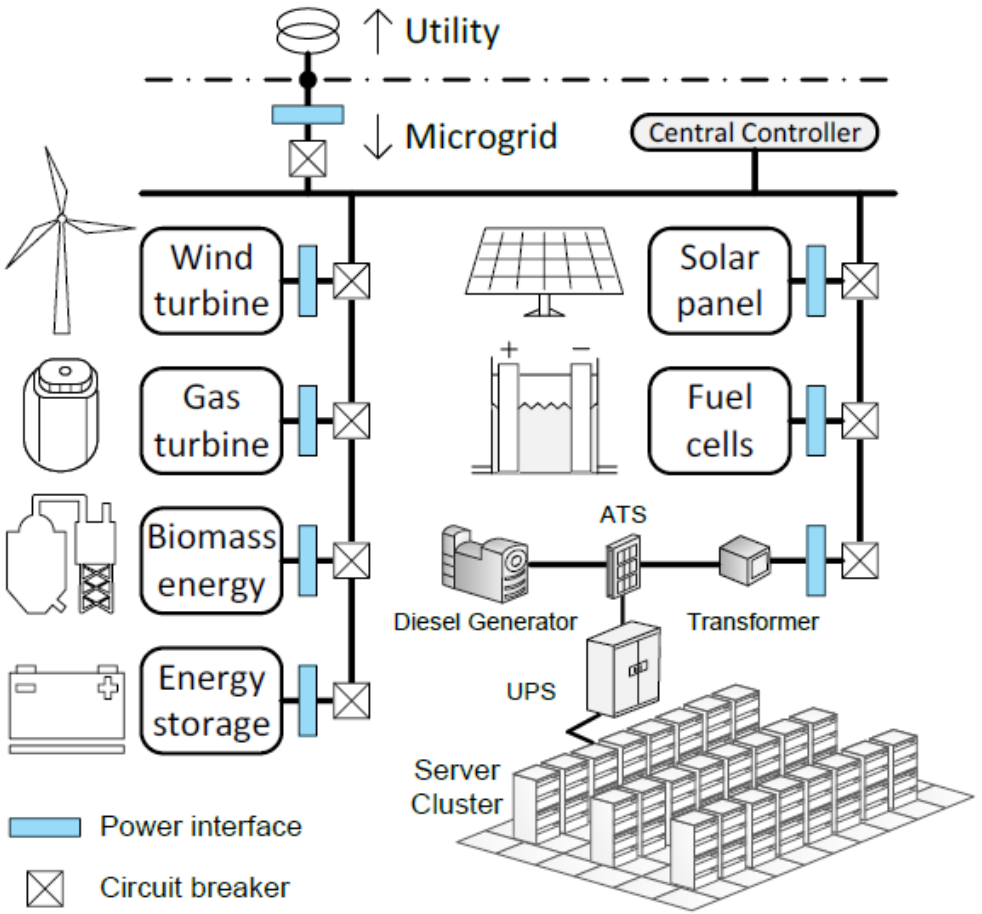
B. Flatten the WSC load profile
   - Use less utility power when it is most expensive



Courtesy of [KAZS12]

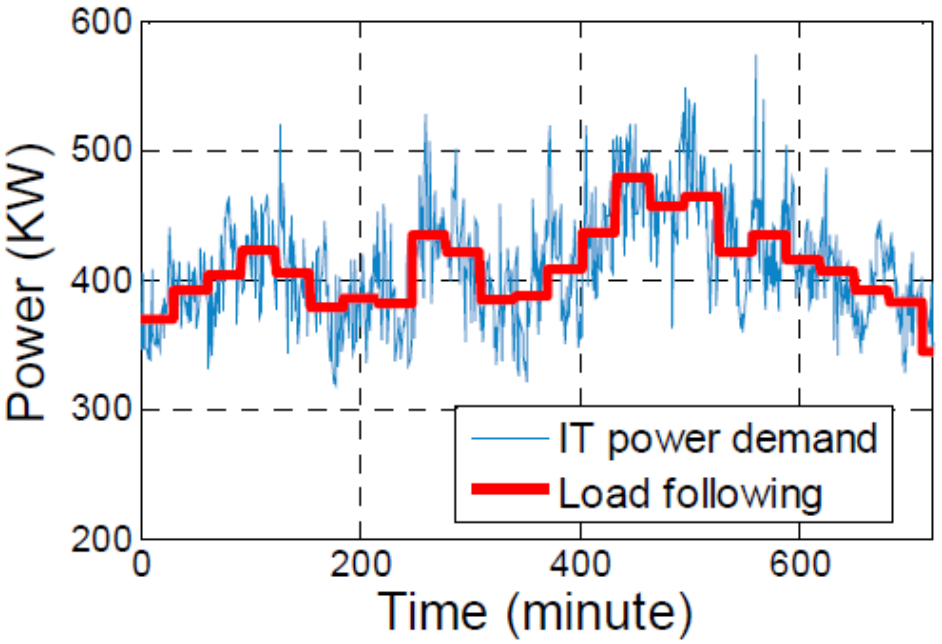
On-site renewable generation

- ↑ Less power transmission and conversion losses
- ↑ Avoid charges imposed by power grid utilities
- ↑ Middleware can assist the power generation
- ↓ Cost to install/maintain the infrastructure
- ↓ Increased complexity of the power supply
- ↓ Suboptimal location of WSC/renewable plant



On-site renewable generation

- Adapt the amount of energy generated to expected workload (i.e. load following)
  - Not adequate when workload fluctuates in short-time intervals
    - Requires some time to adapt the amount of generated energy
  - Combine it with workload shaping



Courtesy of [LiZL13]

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